Solutions, Tenth Annual ECC Undergraduate Mathematics Competition, March 31, 2007

1. Triangular numbers.

If $n = k(k+1)/2$, then

$$9n + 1 = \frac{9k(k+1)}{2} + 1 = \frac{9k^2 + 9k + 2}{2} = \frac{(3k+1)(3k+2)}{2},$$

which is triangular.

2. Sum of the coefficients.

It is $\boxed{2008!}$. For, if $P(x) = a_0 + a_1 x + \cdots + a_n x^n$ is any polynomial, the sum of its coefficients is $a_0 + a_1 + \cdots a_n = P(1)$. For our problem, $P(1) = (1)(2)(3)\cdots(2008) = 2008!$.

3. Sum of integer parts of logs.

If $\boxed{n = 473}$, then $\sum_{k=1}^{n} \lfloor \log_3 k \rfloor = 2007$, as we show now.
For $1 \le k < 3$, $\lfloor \log_3 k \rfloor = 0$.
For $3 \le k < 9$, $\lfloor \log_3 k \rfloor = 1$.
For $9 \le k < 27$, $\lfloor \log_3 k \rfloor = 2$.
For $27 \le k < 81$, $\lfloor \log_3 k \rfloor = 3$.
For $81 \le k < 243$, $\lfloor \log_3 k \rfloor = 4$.
For $243 \le k < 729$, $\lfloor \log_3 k \rfloor = 5$. Thus

$$\begin{aligned}\sum_{k=1}^{242} \lfloor \log_3 k \rfloor &= (9-3)(1) + (27-9)(2) + (81-27)(3) + (243-81)(4) \\ &= 6 + 36 + 162 + 648 = 852,\end{aligned}$$

and $2007 - 852 = 1155 = 231 \cdot 5$, so we need 231 further terms of 5 each. This takes to $n = 242 + 231 = 473$, and $\sum_{k=1}^{473} \lfloor \log_3 k \rfloor = 2007$.

4. Area between circles.

The area is $\boxed{\pi/4}$. If the radius of the inscribed circle is r, then by the Pythagorean Theorem the radius of the circumscribed circle is $\sqrt{r^2 + (1/2)^2}$. Then the area between the circles is

$$\pi(r^2 + 1/4) - \pi r^2 = \pi/4.$$

5. The second largest is 9.

The probability is $\boxed{\frac{7}{33}}$. There are

$$\binom{12}{7} = \frac{(12)(11)(10)(9)(8)}{(5)(4)(3)(2)}$$

ways to choose seven integers from the twelve. The second largest is 9 if and only if one is chosen from $\{9\}$, one from $\{10, 11, 12\}$ and five from $\{1, 2, 3, 4, 5, 6, 7, 8\}$. There are

$$\binom{1}{1}\binom{3}{1}\binom{8}{5} = \frac{(3)(8)(7)(6)}{(3)(2)} = (8)(7)(3)$$

ways to make such a selection. Thus the required probability is

$$\frac{(8)(7)(3)(5)(4)(3)(2)}{(12)(11)(10)(9)(8)} = \frac{7}{33}.$$

6. Equal sums.

If $n = 2m$ is even, we have

$$\begin{aligned}\sum_{k=1}^{n}(-1)^{k+1}k^2 &= (1^2 - 2^2) + (3^2 - 4^2) + \cdots + (2m-1)^2 - (2m)^2 \\ &= (1-2)(1+2) + (3-4)(3+4) + \cdots + (2m-1-2m)(2m-1+2m) \\ &= (-1)(1+2+3+4+\cdots+2m-1+2m) \\ &= (-1)^{n+1}\sum_{k=1}^{2m} k.\end{aligned}$$

If $n = 2m+1$ is odd, then

$$\begin{aligned}\sum_{k=1}^{n}(-1)^{k+1}k^2 &= 1^2 + (3^2 - 2^2) + (5^2 - 4^2) + \cdots + (2m+1)^2 - (2m)^2 \\ &= 1 + (3-2)(3+2) + (5-4)(5+4) + \cdots + (2m+1-2m)(2m+1+2m) \\ &= 1 + (2+3) + (4+5) + \cdots + (2m+2m+1) \\ &= (-1)^{n+1}\sum_{k=1}^{2m+1} k.\end{aligned}$$

Thus, in both cases the identity is established.

(The identity can also be readily proved by induction.)

7. Sum of cosines.

We prove by induction that for every positive integer n,

$$\cos\theta + \cos 3\theta + \cdots + \cos(2n-1)\theta = \frac{\sin 2n\theta}{2\sin\theta}.$$

With $n = 1$ we have the well-known identity $\sin 2\theta = 2\sin\theta\cos\theta$. Assume that k is a positive integer for which

$$\cos\theta + \cos 3\theta + \cdots + \cos(2k-1)\theta = \frac{\sin 2k\theta}{2\sin\theta}.$$

Then

$$\cos\theta + \cos 3\theta + \cdots + \cos(2k-1)\theta + \cos(2k+1)\theta = \frac{\sin 2k\theta}{2\sin\theta} + \cos(2k+1)\theta,$$

so we need to show that

$$\frac{\sin 2k\theta}{2\sin\theta} + \cos(2k+1)\theta = \frac{\sin(2k+2)\theta}{2\sin\theta}.$$

From the identity

$$\sin x - \sin y = 2\cos\frac{1}{2}(x+y)\sin\frac{1}{2}(x-y)$$

we have

$$\sin(2k+2)\theta - \sin 2k\theta = 2\cos(2k+1)\theta\sin\theta,$$

and the desired result follows.

SECOND SOLUTION

We sum the geometric series

$$S(\theta) = e^{i\theta} + e^{3i\theta} + \cdots + e^{(2n-1)i\theta} = \frac{e^{(2n+1)i\theta} - e^{i\theta}}{e^{2i\theta} - 1}.$$

Upon multiplying numerator and denominator by $e^{-2i\theta} - 1$ we obtain

$$S(\theta) = \frac{-e^{(2n+1)i\theta} + e^{(2n-1)i\theta} + e^{i\theta} - e^{-i\theta}}{2 - 2\cos 2\theta}.$$

Now equating real parts we have

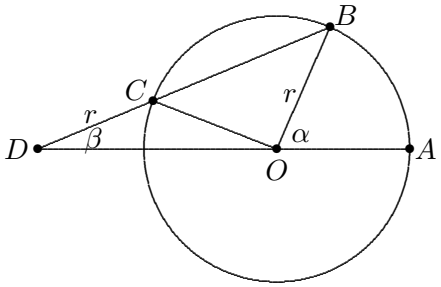
$$\begin{aligned}\cos\theta + \cos 3\theta + \cdots + \cos(2n-1)\theta &= \frac{-\cos(2n+1)\theta + \cos(2n-1)\theta}{2(1-\cos 2\theta)} \\ &= \frac{2\sin 2n\theta\sin\theta}{4\sin^2\theta} = \frac{\sin 2n\theta}{2\sin\theta}.\end{aligned}$$

8. Angle trisection.

Draw the radius OC. Then triangle OCD is isosceles and $\angle COD = \beta$, so $\angle OCD = \pi - 2\beta$. It follows that $\angle OCB = 2\beta$, and because triangle OCB is isosceles, $\angle OBC = 2\beta$ also. Thus $\angle BOC = \pi - 4\beta$, and adding up the angles at O we have

$$\beta + (\pi - 4\beta) + \alpha = \pi,$$

which implies that $\alpha = 3\beta$.



9. A limit.

The limit is $\boxed{\left(1 + \frac{1}{2} + \cdots + \frac{1}{m}\right)a}$. For each integer k, $1 \le k \le m$,

$$\lim_{x\to 0} \frac{1}{x} f\left(\frac{x}{k}\right) = \lim_{x\to 0} \frac{1}{k}\left(\frac{f(x/k) - f(0)}{x/k - 0}\right) = \frac{1}{k} f'(0) = \frac{1}{k} a.$$

Thus

$$\lim_{x\to 0} \frac{1}{x}\left(f(x) + f\left(\frac{x}{2}\right) + \cdots + f\left(\frac{x}{m}\right)\right) = \left(1 + \frac{1}{2} + \cdots + \frac{1}{m}\right)a.$$

10. At least one real root.

They are $\boxed{m = 1 \text{ and } m = 5}$. By substitution one readily verifies that with $m = 1$, the cubic has the real root $z = 1$, and with $m = 5$, $z = -1$ is a root. We must show that there are no other real values of m for which the given cubic has a real root.

Suppose that r is a real root of the cubic equation. Then

$$r^3 + (3 + i)r^2 - 3r - (m + i) = 0,$$

so $r^3 - 3r = -r^2(3 + i) + m + i = (-3r^2 + m) + (-r^2 + 1)i$. Equating real and imaginary parts we find

$$r(r^2 - 3) = r^3 - 3r = -3r^2 + m, \tag{1}$$

and

$$0 = -r^2 + 1. \tag{2}$$

From (2) we see that $r^2 = 1$, so $r = \pm 1$. Substituting into (1) we find $m = 3 - 2r$. Thus $m = 1$ or $m = 5$.